



INSTITUTO SUPERIOR TÉCNICO

DEPARTAMENTO DE ENGENHARIA INFORMÁTICA

COMPUTER ORGANIZATION

LEIC-A, LEIC-T

Third Lab Assignment: Instruction Level Parallelism

Version 2.0

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STUDENTS IDENTIFICATION:

Number:	Name:
ist109881	Tomás Martins Ferreira
ist1110179	Renato Filipe Antunes Neto
ist1109605	Sebastião Gomes Sequeira Lemos de Mendonça

2.1 Simple execution, without data forwarding techniques

e)	Clock cycles	308		Stalls: - Data	104
	Instructions	170		- Structural	0
	Average CPI	1.81		Flushes: - Branch Taken	15

f) A política de previsão de branch utilizada aqui é “branch not taken”, o que significa que o programa continua a executar de forma sequencial após um branch. Se o branch revelar ser um salto, o pipeline é esvaziado (flush) e as instruções corretas são carregadas. É possível ver isso no diagrama no pipeline, uma vez que após a instrução de branch as instruções addi e ecall começam a ser executadas e param de ser executadas ao saber a decisão de salto do brach.

2.2 Application of data forwarding techniques

c)

Clock cycles	222
Instructions	170
Average CPI	1.31

Stalls: - Data	18
- Structural	0
Flushes: - Branch Taken	15

d)

$SpeedUp = T(old) / T(new)$. Como o tempo de clock é sempre igual, podemos dizer que $SpeedUp = Cycles(old) / Cycles(new) = 308 / 222 = 1.39$

2.3 Source code optimization: minimization of data and structural hazards

a) Attach a copy of the new assembly program.

```
.data
A: .word 1,2,3,4,5,6,7,8,9
   .word 10,11,12,13,14,15,16
B: .word 11,22,33,44,55,66,77
   .word 88,99,111,122,133
   .word 144,155,166
C: .word 0,0,0,0,0,0,0,0,0
   .word 0,0,0,0,0,0,0
.text
main:
    addi s1, zero, 0 # i = 0
    addi s2, zero, 16 # value of N
    la s3, A # base of A
    la s4, B # base of B
    la s5, C # base of C
loop:
    lw t1, 0(s3)
    lw t2, 0(s4)
    addi s1, s1, 1
    addi s3, s3, 4
    addi s4, s4, 4
    mul t3, t2, t1
    add t3, t3, t1
    sw t3, 0(s5)
    addi s5, s5, 4
    bne s1, s2, loop
end:
    addi a7, zero, 10
    ecall # Exit (syscall)
```

c)

Clock cycles	206
Instructions	170
Average CPI	1.21

Stalls: - Data	2
- Structural	0
Flushes: - Branch Taken	15

d)

$$\text{SpeedUp} = T(\text{old})/T(\text{new}) = \text{Cycles}(\text{old})/\text{Cycles}(\text{new}) = 308/206 = 1.50$$

2.4 Source code optimization: loop unrolling

a) Attach a copy of the new assembly program.

```
.data
A: .word 1,2,3,4,5,6,7,8,9
   .word 10,11,12,13,14,15,16
B: .word 11,22,33,44,55,66,77
   .word 88,99,111,122,133
   .word 144,155,166
C: .word 0,0,0,0,0,0,0,0,0
   .word 0,0,0,0,0,0,0
.text
main:
    addi s1, zero, 0 # i = 0
    addi s2, zero, 16 # value of N
    la s3, A # base of A
    la s4, B # base of B
    la s5, C # base of C
loop:
    lw t1, 0(s3)
    lw t2, 0(s4)
    addi s1, s1, 1
    mul t3, t2, t1
    add t3, t3, t1
    sw t3, 0(s5)

    lw t1, 4(s3)
    lw t2, 4(s4)
    addi s1, s1, 1
    mul t3, t2, t1
    add t3, t3, t1
    sw t3, 4(s5)

    lw t1, 8(s3)
    lw t2, 8(s4)
    addi s1, s1, 1
    mul t3, t2, t1
    add t3, t3, t1
    sw t3, 8(s5)

    lw t1, 12(s3)
    lw t2, 12(s4)
    addi s1, s1, 1
    mul t3, t2, t1
    add t3, t3, t1
    sw t3, 12(s5)

    addi s3, s3, 16
    addi s4, s4, 16
    addi s5, s5, 16

    bne s1, s2, loop
end:
    addi a7, zero, 10
    ecall # Exit (syscall)
```

c)

Clock cycles	134
Instructions	122
Average CPI	1.1

Stalls: - Data	2
- Structural	0
Flushes: - Branch Taken	3

$$\text{SpeedUp} = T(\text{old})/T(\text{new}) = \text{Cycles}(\text{old})/\text{Cycles}(\text{new}) = 308/134 = 2.30$$

	INSTRUCTIONS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	
1	lw x6 0 x19	F	D	E	M	W														F																						
2	lw x7 0 x20		F	D	E	M	W																																			
3	mul x28 x7 x6			F	D	-	-	E	M	W																																
4	add x28 x28 x6				F	-	-	D	-	-	E	M	W																													
5	sw x28 0 x21							F	-	-	D	-	-	E	M	W																										
6	addi x9 x9 1										F	-	-	D	E	M	W																									
7	addi x19 x19 4													F	D	E	M	W																								
8	addi x20 x20 4														F	D	E	M	W																							
9	addi x21 x21 4															F	D	E	M	W																						
10	bne x9 x18 -36																F	D	E	M	W																					
11	addi x17 x0 10																	F	D																							
12	ecall																		F																							
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Table 1: Pipeline time diagram, without data forwarding techniques

	INSTRUCTIONS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
1	lw x6 0 x19	F	D	E	M	W									F																										
2	lw x7 0 x20		F	D	E	M	W																																		
3	mul x28 x7 x6			F	D	-	E	M	W																																
4	add x28 x28 x6				F	-	D	E	M	W																															
5	sw x28 0 x21						F	D	E	M	W																														
6	addi x9 x9 1							F	D	E	M	W																													
7	addi x19 x19 4								F	D	E	M	W																												
8	addi x20 x20 4									F	D	E	M	W																											
9	addi x21 x21 4										F	D	E	M	W																										
10	bne x9 x18 -36											F	D	E	M	W																									
11	addi x17 x0 10												F	D																											
12	ecall													F																											
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Table 2: Pipeline time diagram, with data forwarding techniques.

[illegible]

Table 3: Pipeline time diagram, with minimization techniques to reduce the data and structural hazards.

	INSTRUCTIONS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
1	lw x6 0 x19	F	D	E	M	W																										F									
2	lw x7 0 x20		F	D	E	M	W																																		
3	addi x9 x9 1			F	D	E	M	W																																	
4	mul x28 x7 x6				F	D	E	M	W																																
5	add x28 x28 x6					F	D	E	M	W																															
6	sw x28 0 x21						F	D	E	M	W																														
7	lw x6 4 x19						F	D	E	M	W																														
8	lw x7 4 x20							F	D	E	M	W																													
9	addi x9 x9 1								F	D	E	M	W																												
10	mul x28 x7 x6									F	D	E	M	W																											
11	add x28 x28 x6										F	D	E	M	W																										
12	sw x28 4 x21											F	D	E	M	W																									
13	lw x6 8 x19												F	D	E	M	W																								
14	lw x7 8 x20													F	D	E	M	W																							
15	addi x9 x9 1														F	D	E	M	W																						
16	mul x28 x7 x6															F	D	E	M	W																					
17	add x28 x28 x6																F	D	E	M	W																				
18	sw x28 8 x21																	F	D	E	M	W																			
19	lw x6 12 x19																		F	D	E	M	W																		
20	lw x7 12 x20																			F	D	E	M	W																	
21	addi x9 x9 1																				F	D	E	M	W																
22	mul x28 x7 x6																					F	D	E	M	W															
23	add x28 x28 x6																						F	D	E	M	W														
24	sw x28 12 x21																							F	D	E	M	W													
25	addi x19 x19 16																								F	D	E	M	W												
26	addi x20 x20 16																									F	D	E	M	W											
27	addi x21 x21 16																										F	D	E	M	W										
28	bne x9 x18 -108																											F	D	E	M	W									
29	addi x17 x0 10																												F	D											
30	ecall																														F										

Table 4: Pipeline time diagram: usage of loop unrolling minimization techniques to reduce the control hazards.